

Homework 3

Submission 2, Spring 2026

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Problem 1

```
# A tibble: 6 x 5
  year `Mean Star Rating` `Mean Enrollments` `Mean Market Share` `Total Plans`
  <dbl>          <dbl>          <dbl>          <dbl>          <int>
1  2010              2.97            256.            0.0658        108222
2  2011              3.26            343.            0.0855         68003
3  2012              3.36            377.            0.0856         67246
4  2013              3.55            396.            0.0815         67835
5  2014              3.81            437.             0.08         62344
6  2015              3.96            470.            0.0787         65499
```

Problem 2

```
# A tibble: 6 x 4
  year `Mean Enrollments` `Mean Market Share` `Total Plans`
  <dbl>         <dbl>         <dbl>         <int>
1  2010             107.             0.0673         48643
2  2011             168.             0.09           12754
3  2012             173.             0.0567          9077
4  2013             214.             0.06           3661
5  2014             258.             0.0557          3708
6  2015             314.             0.0405          4680
```

Problem 3

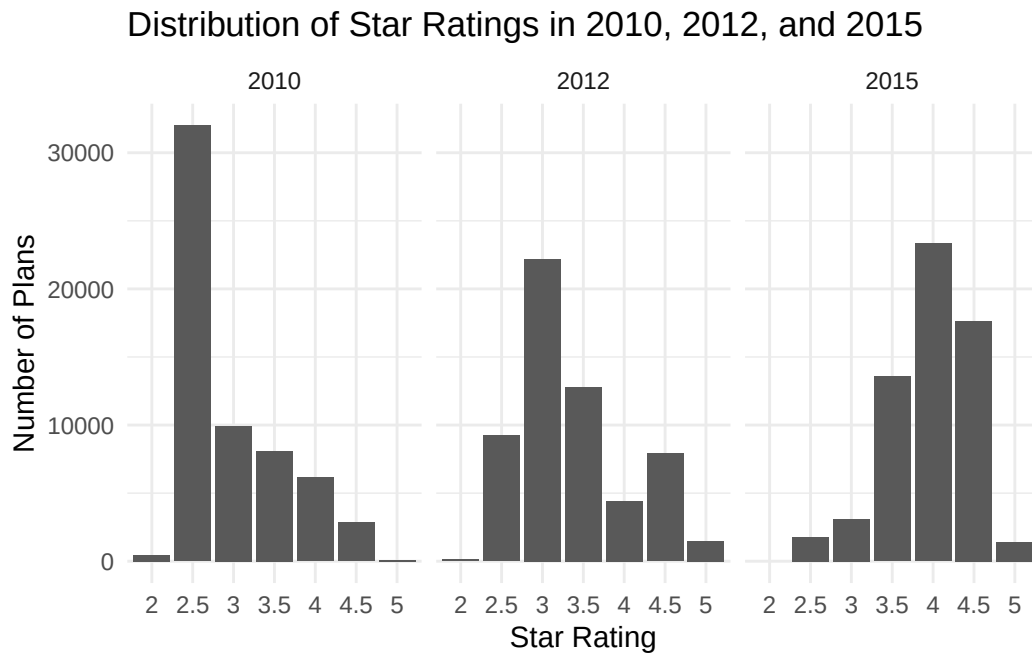


Figure 1: Distribution of Star Ratings (2010, 2012, 2015)

Problem 4

	2010	2011	2012	2013	2014	2015
Intercept (2.5 stars)	0.0634	0.0702	0.0789	0.0491	0.0640	0.0606
3 Stars	0.0126	0.0350	0.0307	0.0621	0.0276	0.0271
3.5 Stars	0.0005	0.0191	0.0129	0.0385	0.0274	0.0388
4 Stars	-0.0036	0.0028	-0.0246	0.0172	0.0074	0.0115
4.5+ Stars	-0.0076	-0.0095	-0.0134	0.0222	0.0071	0.0233

Problem 5

```
# A tibble: 5 x 2
  `Star Rating`    `Number Of Plans`
  <chr>           <int>
1 Rounded up to 3      1978
2 Rounded up to 3.5    2073
3 Rounded up to 4     1056
4 Rounded up to 4.5     150
5 Rounded up to 5        15
```

Problem 6

Statistic	3 Stars	3.5 Stars
Intercept	0.016	0.102
Rounded	0.055	-0.040
Running Score	-0.740	0.417
Num.Obs.	5309.000	1882.000
R2	0.059	0.015

Problem 7

3 stars estimates are stable across all bandwidths, 3.5 estimates are more sensitive to bandwidth choice. Both are relatively robust as the sign of both effects do not change.

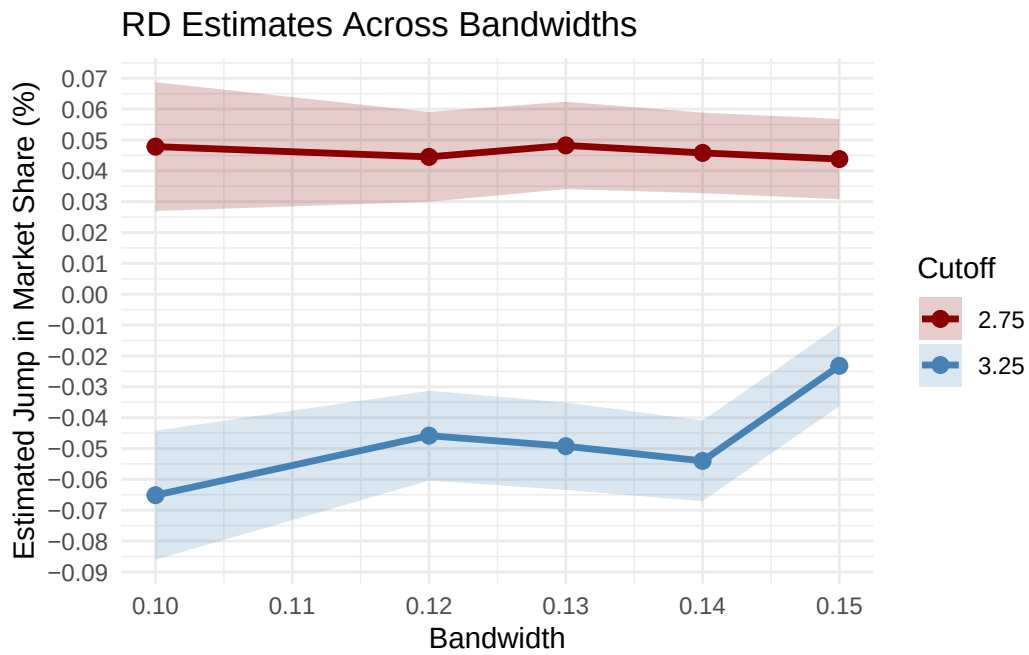


Figure 2: Different Bandwidths

Problem 8

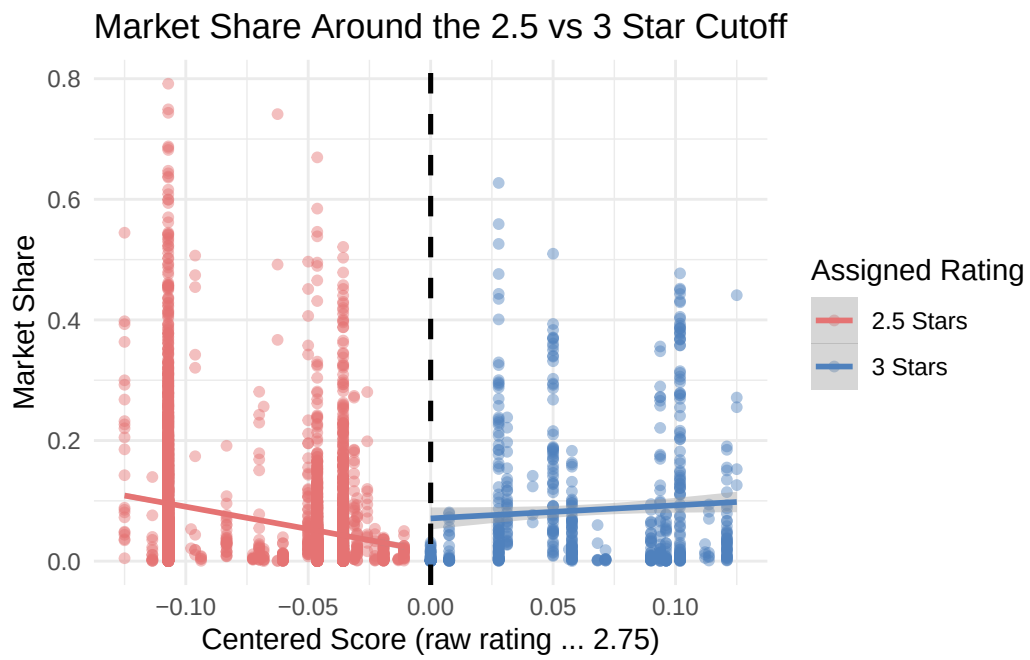


Figure 3: Distribution of Running Rariable Near 2.75 Threshold

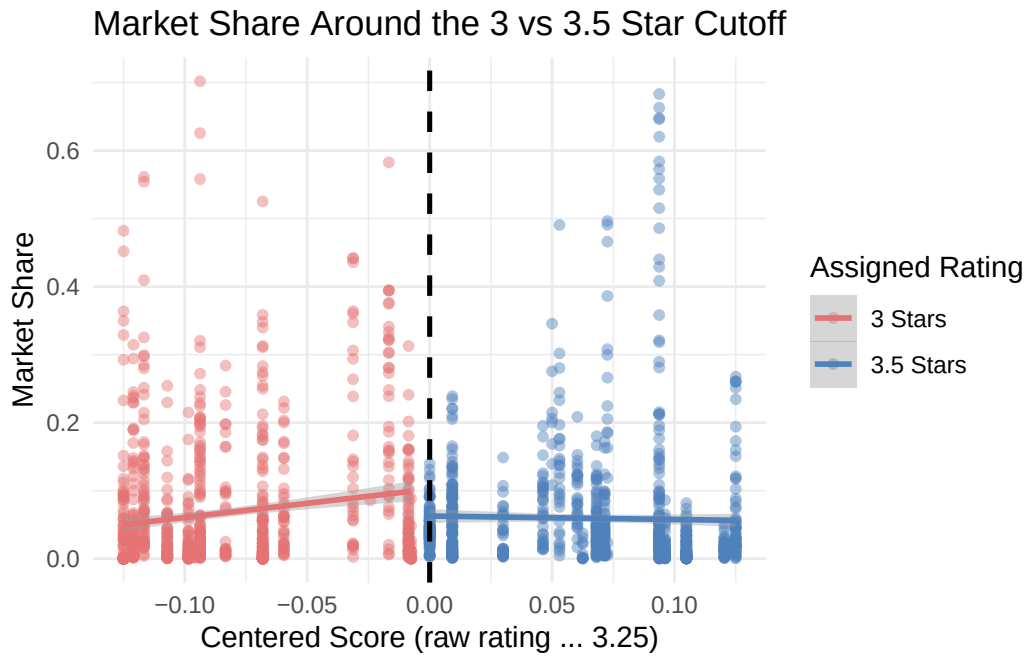


Figure 4: Distribution of Running Rariable Near 3.25 Threshold

Problem 9

Table: Plan Characteristics Around the 2.75 Cutoff

```
# A tibble: 2 x 4
  group      share_HMO share_PartD     n
  <chr>      <dbl>      <dbl> <int>
1 Above cutoff    0.447        0.840  2528
2 Below cutoff    0.0194       0.549 22385
```

Table: Plan Characteristics Around the 3.25 Cutoff (3-3.5 Stars)

```
# A tibble: 2 x 4
  group      share_HMO share_PartD     n
  <chr>      <dbl>      <dbl> <int>
1 Above cutoff    0.377        0.817  2452
2 Below cutoff    0.606        0.864  3969
```

These tables show you whether plans just above each cutoff look similar to plans just below it. Plans just above the 2.75 threshold are completely different types of plans as the ones above are much more likely to be HMOs and to offer Part D. Plans for just above the 3.25 threshold are slightly different but to a lesser degree.

Problem 10

Overall, a higher star rating shows more enrollment but the comparison is compromised as plans above and below the threshold are different. Raw ratings tend to be more distributed around the thresholds (2.75 and 3.25), and the adjusted ratings create jumps in star assignments. Because plan characteristics are distinct and not comparable right above or below the cutoff, the RD model cannot cleanly isolate the effect of the star rating alone.